

Reading Circle

Towards a More In-Depth Detection of Political Framing

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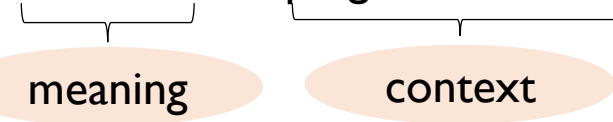
Paper Title: Towards a More In-Depth Detection of Political Framing

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Research problem:

Current NLP approaches **largely focus on detecting what topics are emphasized** (e.g., economy or security) but ignore how the language itself subtly shapes a reader's perception through semantic and pragmatic devices.



Motivation:

Framing is not just about selecting a topic; **linguistic choices can heavily influence interpretation** without changing the underlying factual content.

(1) *Die drei Parteien wissen ja nicht einmal, was im Falle der Flüchtlinge [...] unter Familie zu verstehen ist.*

‘The three (ruling) parties do **not even** understand what family means in the case of the refugees - **as we all know**.’

(source: *Frankfurter Allgemeine Zeitung*)

“**not even**” strengthens the author’s **criticism**, implying that the **ruling parties lack even the most basic understanding** of family in asylum cases.

“**As we all know**” presents the author’s opinion as **shared knowledge**. This makes the opinion seem **widely accepted** and increases its credibility.

Main Idea:

Investigating which linguistic structures construct the framing: Only the topics or something else?

Practical side

Modeling side

Existing Works

Supervised approach

Neural approaches

Lack of explainability

Intense manual
annotation effort

For Other Lang:
Not found

For English Lang:
Many datasets available

Unsupervised approach

Topic models based on
compiled framing vocabulary

Supervised approach

Statistical
learning method

Naïve Bayes:
Theoretically driven linguistic
features

Topic oriented

Neural approaches

Recurrent NN

Long Short Term
Memory Nets

Transformer
based LM

We know very little about Linguistic Properties of Framing

Topic oriented

What is “Framing”?

- Framing is a strategy to promote certain views, values, or ideologies in political discourse.
- Information sender selectively makes certain aspects of an issue more noticeable in the discourse while excluding or denying the others.

Data:

Newspaper articles about “Refugees” were collected from **BILD, FAZ, and SZ** from 2014 to 2018.

Topical Framing:

About what topics are covered Used “Refugee and Migration Frame Vocabulary (RMFV) with 9 distinct topics.

Economy, Identity, Legal, Morality, Policy, Politics, Public Opinion, Security and Welfare

$$\text{Topical Ratio} = \frac{\text{Frequency of vocabulary words from a specific frame (e.g., Security)}}{\text{Total tokens in an article}}$$

Rhetorical Framing

Arousal:

Framing an issue with emotionally charged language that can make persuasive impact on the addressee.

Presupposition:

Using scalar particles like even / not even, adverbs like again / continuously for iteration or continuation. This emphasize that something is a longstanding problem and framed as criticism.

- (5) *Das seit Monaten andauernde unwürdige Chaos bei der Aufnahme von Flüchtlingen in Berlin hat zu personellen Konsequenzen geführt.* (BILD)
 ‘The continuous undignified chaos in refugee reception in Berlin has led to personnel consequences.’

Sentence Type:

Utilize rhetorical questions to keep the addressee engaged in an interactive style. Framing as evident, covertly prevents reader from questioning.

- (4) *Viele Flüchtlinge wollen gar nicht in den Turnhallen wohnen!* (BILD)
 ‘Many refugees don’t want to be accommodated in the gym!’

Modal Particles: No English equivalent

Common ground – Framing something as a widely accepted knowledge.(eg: eben – obviously)

Resigned – Speaker believes something obvious and cannot be altered therefore has to be accepted.

Weak commitment- Speaker consider that something is highly probable but not 100% certain.

Information Structure:

Organization of how sentences are inter-related
 Usage of yet, because, though, if.

To make argument strong and rational.

- (8) *Im Mittelmeer wird derweil weiter gestorben – weil es eben für Flüchtlinge und Migranten keinen legalen Weg nach Europa gibt.* (SZ)
 ‘Meanwhile, there are further death cases at the mediterranean sea – **because** there is **eben** no legal route to Europe for refugees and migrants.’

Methodology and Results

Computing most predictive features using z – value

Binary Logistic Regression model was fitted and its coefficient of estimate is obtained.

$$z - value_i = \frac{Coefficient\ Estimate\ (\beta_i)}{Standard\ Error\ (SE_i)}$$

BILD uses **strong language** through **exclamatory markers** regarding **security issues** more by **less focusing on morality issues** framing **refugees** as a **threat**.

On the other hand,

FAZ and **SZ** frame the issue in a **more humanitarian manner** by focusing more on **morality** with using a **more logical informational structures** using **connectives**.

SZ use more **emotionally charged languages** as its z value for “**Arousal degree**” is higher than **FAZ**.

Positive z-value (+): The feature is predictive for articles **belonging to the source**. For example, the **SECURITY** frame has a z-value of +14.041 for **BILD**, meaning articles with **SECURITY related coverage are much more likely** to be from **BILD**

Negative z-value (-): The feature is predictive for articles **not belonging to the source**. For example, the **SECURITY** frame has a z-value of -10.921 for **FAZ**, meaning **FAZ articles systematically use this frame less** than the other newspapers in the study.

Significance of each predictor is determined by the computed **z-value**

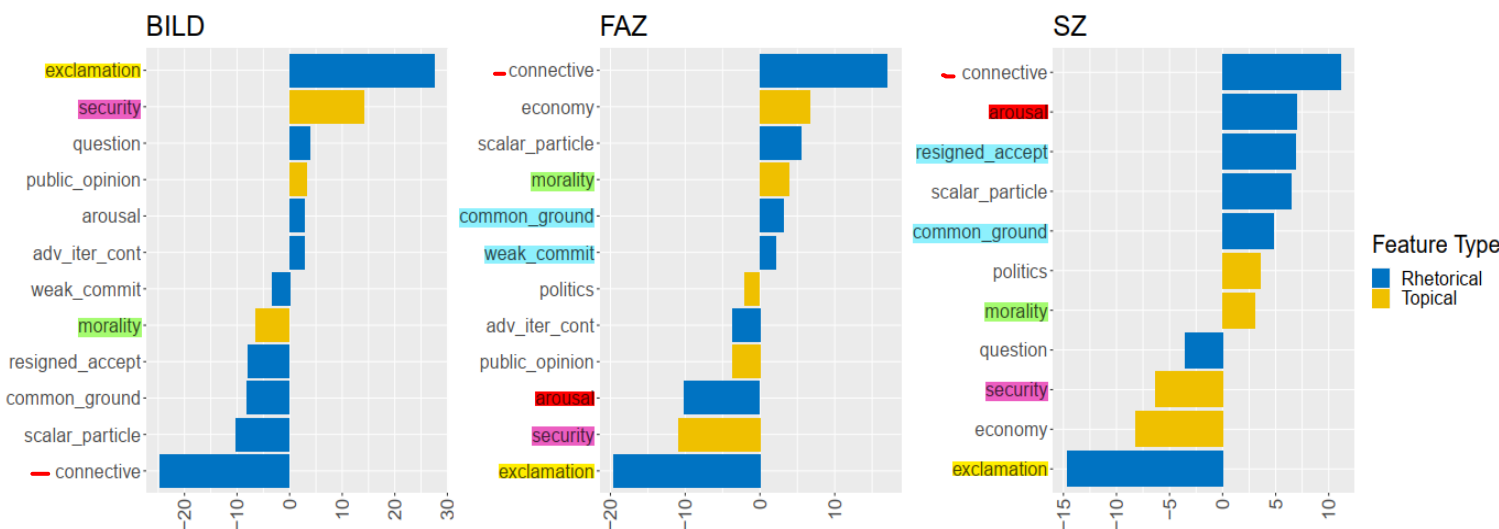


Figure 3: Z-values of each feature for BILD-, FAZ-, and SZ-articles. Only significant features are shown.

What they are trying to do actually?

- Based on a set of features, they predict **which source (newspaper) a particular news article belongs to**.

What they are not trying to do?

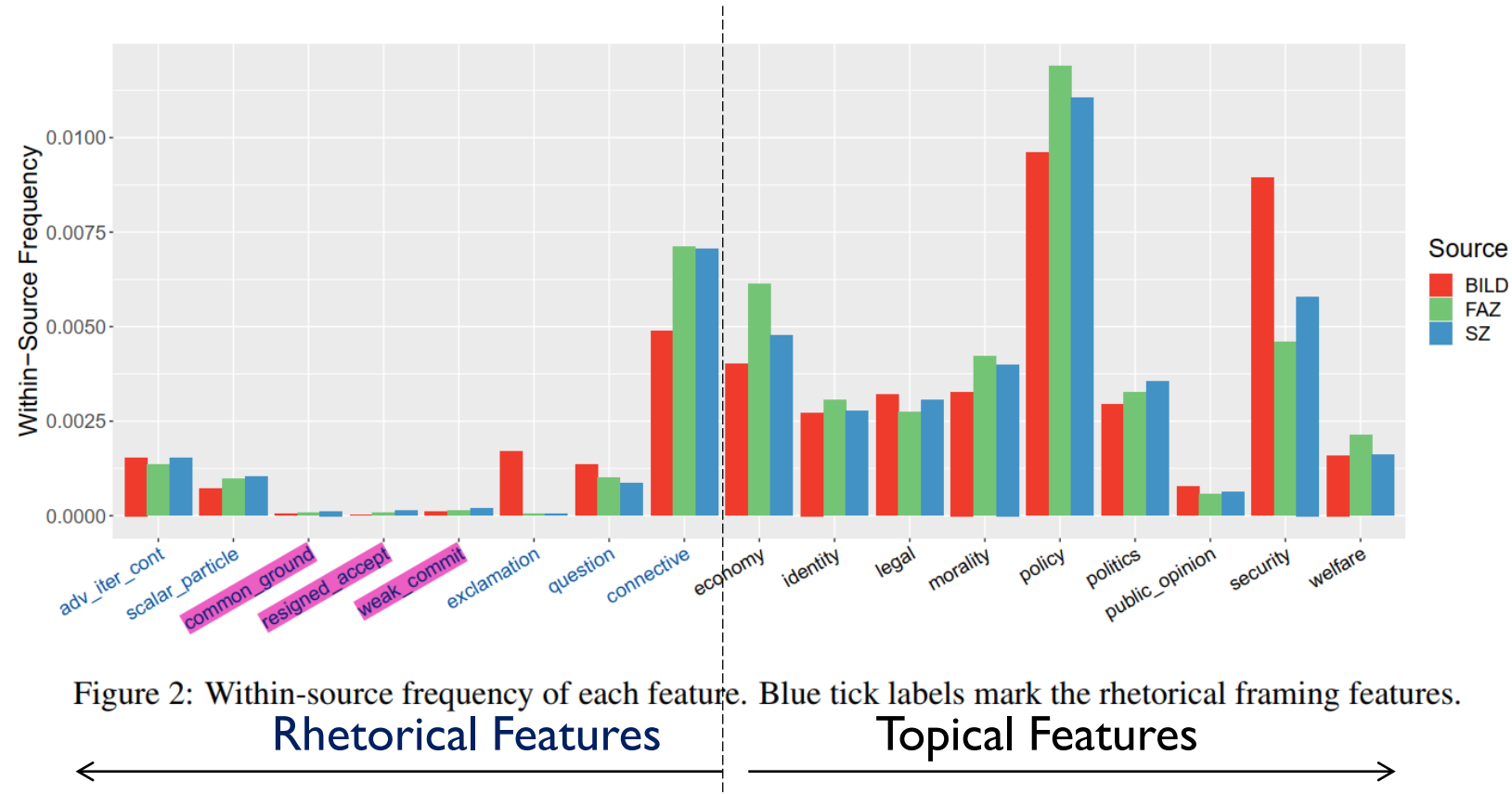
- They **do not predict** which **single frame** (e.g., SECURITY / MORALITY) an article belongs to.
- The highest “Topical Ratio” shows which frame is emphasized most in an article, but it does not mean that this is the only frame discussed.

Purpose of doing this?

- To identify the specific framing strategies that distinguish one paper from another
 - BILD-articles are best predicted by "exclamation marks" and "security topic"
 - FAZ-articles are predicted by the use of "connectives" and "economy topic"

What we can learn from this paper?

- Hand-crafted features are important to consider.
- The frequency of modal particles are sparse compared to other features, difficult for a standard neural network to capture this.
- Framing should not be just limited to topic coverage. Rhetorical Framing also has to be considered.



- For the same event, “**The European Refugee Crisis,**” they used the “**Topical**” and “**Rhetorical Marker**” profiles of each newspaper to show how different sources use different strategies to frame the event.
- This research is designed specifically for German language (i.e: Modal particles found in German are not present in all other languages). But the research method can be adopted.

Thank you